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Geochemistry of Darwin impact glass and target rocks

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Abstract—We have analyzed the major and trace element composition of 18 Darwin glass samples and 7 target rocks (sandstones, shales, and a quartzite) from the Darwin crater area. On the basis of our data, and using statistical methods, 3 chemically distinct groups of Darwin glass were identified: A (low Fe, Al = LFe,Al, or average Darwin glass group), B (HFe,Al group), and C (HMg,Na group). The glasses of group C also show anomalous enrichments of several elements, e.g., Cr, Mn, Co, and Ni. Electron microprobe studies show that the glasses are inhomogeneous on the micrometer scale, which is typical for impact glasses. The geochemistry of all glasses is very similar to terrestrial sediments and thus supports the impact origin model. We have performed mixing calculations which show that in general Darwin glass can be formed by melting and mixing local target rocks. The best fit is obtained for a mixture of 30% quartzite, 60% shale B3-DG, and 10% shale B1-DG. Some major element contents do not agree exactly, which is most probably due to the limited selection of target rocks that were available for our study. The analyses and mixing models demonstrate that volatile elements (e.g., Zn, Ga, Sb, and the alkalies) have been lost during production of the impact glasses, which can be expected because of the high formation temperature. We have furthermore tried to explain the enrichments of Cr, Mn, Co, and Ni in group C glasses by contributions from a non-sedimentary source, e.g., ultrabasic rocks, or from the impacting body. None of the mixtures provides a satisfactory fit. Darwin glass does not show any significant Ir enrichments. Admixture of material from iron meteorites gives too high Fe, Co, and Ni, and too low Cr and Mn contents. Chondritic contaminations would yield Ir abundances in the glass that are several orders of magnitude above the observed levels. Better fits are obtained for an achondritic contamination, but again give excess Ir. An ultrabasic contribution gives better results, except for higher Mg, but no such rocks are known from the target area. Thus, at the present time, we are not able to explain the enrichments of Cr, Mn, Co, and Ni in glasses of group C in a satisfactory way.

1. INTRODUCTION

DARWIN GLASS HAD BEEN known by locals for a long time before it was first described and analyzed by SUESS (1914). The first area from which the glass was reported was a locality called Ten Mile Hill in the vicinity of Mt. Darwin, about 20 km south of Queenstown, Tasmania, Australia. SUESS (1914) classified the glass as a new type of tektite. Further analyses have been reported by DAVID et al. (1927); SPENCER (1933, 1939); PREUSS (1935); SUESS (1935); EHMANN (1960); and KOEBERL et al. (1984a,b, 1985, 1986). Darwin glass was classified by these authors as either a tektite or as impact glass formed by fusion of silicate sediments by meteorite impact.

Coesite and tourmaline in Darwin glass were described by REID and COHEN (1962) and thus provided evidence that only terrestrial material could have been the source of the glass. A major geochemical study of Darwin glass was made by TAYLOR and SOLOMON (1964). They analyzed major and trace elements in seven glass samples and several country rocks from Ten Mile Hill and concluded that: 1) Darwin glass was not produced by a terrestrial igneous event; 2) the chemical composition of the glass resembles terrestrial sediments, most likely an argillaceous sandstone; 3) the Cr/Ni, Ni/Co, and Fe/Ni ratios, and high Ni abundances are anomalous for terrestrial rocks; 4) at least two groups of Darwin glass can be distinguished through chemical differences in trace elements; 5) Darwin glass is not related to australites; and 6) the geochemistry of the glass is consistent with a terrestrial origin by meteorite impact. ZÄHRINGER and GENTNER (1963) showed that the Ar-isotope ratios in bubble-rich glasses are similar to the terrestrial atmosphere, adding another argument to the case for a terrestrial impact origin.

TAYLOR and EPSTEIN (1969) report $^{18}O/^{16}O$ values, which are characteristic for terrestrial sandstones, shales, and most other sedimentary rocks. They also demonstrated that oxygen isotope ratios of Darwin glass are different from $^{18}O/^{16}O$ ratios of australites and that a common origin of these two natural glasses can be excluded. MATSUDA and YAJIMA (1989) measured excess Ne in Darwin glass compared to Ne abundances in the present atmosphere and explained this enrichment by diffusion of Ne from the atmosphere into the glass. Ne diffuses easier into the glass than Ar does; thus, higher Ne/Ar ratios are obtained.

The age of the glass has been determined by the K/Ar-method to be 0.73 ± 0.04 Ma (GENTNER et al., 1973). STORZER and WAGNER (1980a,b) reported a fission track age of Darwin glass of 0.81 ± 0.04 Ma, while for australites an age of 0.82 ± 0.05 Ma was determined. The fission track age measurements put the ages of these two natural glasses close

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to each other, so a connection between these two events (e.g., simultaneous impacts from a body that disintegrated before entering the atmosphere) could be considered.

The geochemical data presented by TAYLOR and SOLOMON (1964) argue against a lunar origin of Darwin glass and favor terrestrial parent materials as precursor of the glass. Previously the absence of an impact crater associated with the glass provided problems for the impact theory, but in 1972 R. J. Ford found a crater-like structure near Mt. Darwin (FORD, 1972).

The Darwin crater, which was suggested to be the source crater of Darwin glass, was described by FORD (1972) and FUDALI and FORD (1979). The structure is situated 26 km SSE of Queenstown, at the eastern boundary of the strewnfield, which has been estimated to extend over 400 km^2 (FUDALI and FORD, 1979). The area is heavily vegetated and outcrops of country rocks are very rare; thus, a detailed geological investigation is difficult. The structure is situated in a series of lightly metamorphosed Silurian and Devonian slates, argillites, and faulted and disrupted quartzites (FUDALI and FORD, 1979). Typical features associated with impact craters, such as shocked quartz, an elevated rim, or shatter cones, have not been described in the literature. Although we are in disagreement as to whether or not the evidence for impact origin of the structure is compelling, we will refer to it herein as the Darwin crater.

The aim of this study was to analyze major and trace elements in Darwin glass and the outcropping target rocks in order to establish a geochemical relationship between the impact glass and its parent material.

2. SAMPLE DESCRIPTION

Darwin glass is a natural glass of variable shape and size. It can be found as fragments in the top soil cover, but especially on the gravel road and road cuts which have been washed out by rain. It occurs in fragments ranging from 10 mg to several hundred grams (FUDALI and FORD, 1979). The glass is usually compact with few vesicles, but sometimes is of frothy appearance. The color varies from pitchblack to bottlegreen and almost colorless (translucent). It shows flow structures, which are, however, less pronounced than in other impact glasses (BARNES, 1963). Lechatelierite is common and often has a frothy and vesicular structure. Figure 1 shows two different Darwin glass samples to demonstrate the different shapes and colors.

We have analyzed 18 Darwin glass samples for major and trace elements in order to establish a complete geochemical database for comparison with target rocks. The glass specimens (DG8701 to DG8718) had different shapes, colors, and sizes and weights between 0.64 to 5.70 g. Most of them were of dark color ranging from black to olive green, but a few were light green and translucent. Bubbles with sizes of up to 5 mm diameter were frequently observed. Frothy white parts were found together with denser (almost vesicular free) parts. Some bubbles were stretched—probably by viscous glass flow during cooling—while in other samples no deformation was observed.

Two thin sections of Darwin glass samples are shown in Fig. 2. They clearly display stress and strain features, which are commonly observed in tektites and impact glasses

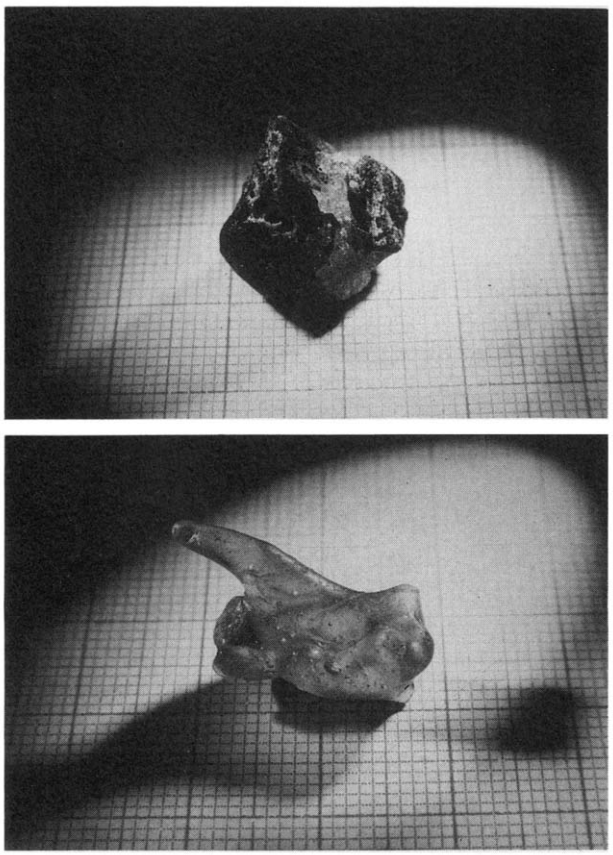


FIG. 1. Photographs showing typical Darwin glass specimens with characteristic shapes and colors: (a) dark glass with lighter colored inclusion of frothy glass; (b) a translucent and abraded specimen with flow features. (The grid in the pictures is in mm.)

(BARNES, 1963). The internal structure of the glass is marked by differences of the R.I. and the color of schlieren. Some samples contain greenish layers that are about 0.1 mm (or less) in thickness and extend over variable lengths (up to a few cm). The possibility of a correlation between color and chemical variations was studied by electron probe microanalysis.

Three shales, B1 to B3-DG, three sandstones, C1 to C3-DG, and one quartzite, A-DG, country rocks were exposed and collected in the vicinity of the crater by one of us (RJF).

3. ANALYTICAL METHODS

Sample preparation. The glass samples were cleaned ultrasonically in distilled water, and then crushed in an agate mortar and powdered in an automatic agate ball mill. 50 to 200 mg of the sample powder were used for instrumental neutron activation analysis (INAA), and about the same amount was used for the spectrometric analyses. Thin sections of representative glass specimens were prepared to investigate the internal structure of the glass with optical and electron microscopes.

Major elements. The contents of Al, Fe, Mg, Ca, and Ti in bulk samples were determined by direct current plasma spectroscopy (DCP), using a Spectraspan IIIB instrument. Solutions of glass and target rock samples were obtained by dissolution of the sample powders in a H_2SO_4/HF acid mixture in platinum crucibles. Prior to dissolution, the target rock powders were heated for 12 h at 110°C, and afterwards for 1 h at 900°C to determine the water content, and L.O.I., respectively. Potassium was determined by atomic absorption

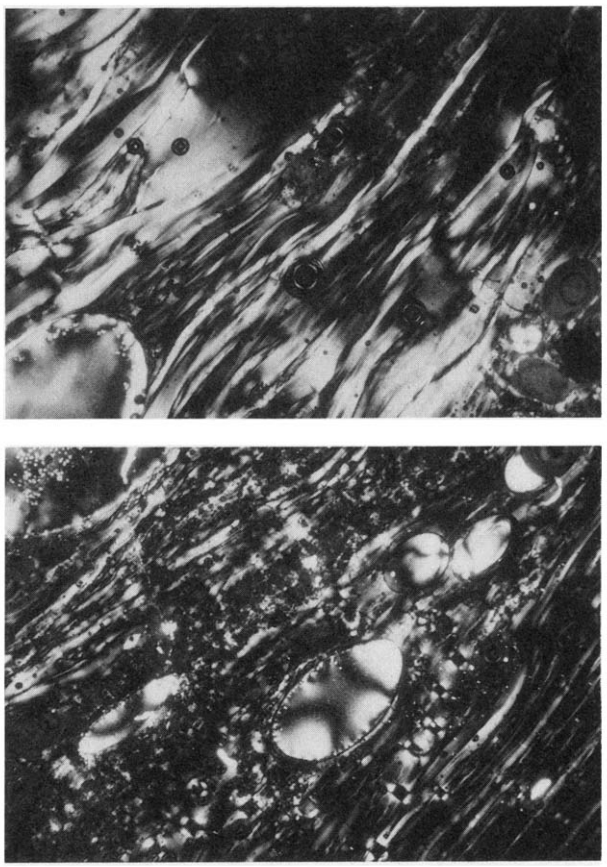


FIG. 2. (a) Thin section of a Darwin glass showing layering, stress, and flow features. (b) The thin section shows large elongated vesicles (flow structure) and layering (both pictures: crossed nicols, picture size: 1.66 × 1.11 mm).

spectrometry (AAS) using a Perkin Elmer AA spectrometer model 303. Sodium was analyzed by INAA. Glass chips of all 18 glass samples, which were also analyzed for trace elements, and an additional sample (collected in 1988 by TM) were analyzed by electron microprobe analysis (EPMA) using a fully computerized 5-spectrometer ARL-SEMQ electron microprobe for Si, Al, Fe, Mg, Ca, K, and Ti.

Trace elements. Sc, Cr, Mn, Co, Ga, As, Rb, Zr, Sb, Cs, Ba, La, Ce, Nd, Sm, Eu, Tb, Dy, Yb, Lu, Hf, Ta, Th, and U were determined by INAA. The analytical accuracy for these elements was checked by analyzing BCR-1 and other natural standards and is generally <5–10%, except for Zr and Sb. The final data have been corrected using the certified natural standard contents (GOVINDARAJU, 1984). The contents of Ni, Cr, and Mn in all samples and Cu, Zn in the sedimentary target rocks were measured by DCP-spectrometry. Li, Be, Cu, Zn, Cd, and Pb in Darwin glass were determined by graphite furnace atomic absorption spectrometry, using Perkin Elmer Model 3030, HGA 400, and AS-1 instrumentation.

4. RESULTS

4.1. *Darwin glass*

Results for major and trace elements are given in Table 1. Silica data was taken from EPMA. The major elements show large intersample variations. The abundances of SiO_2 and TiO_2 show a smaller range (SiO_2 varies from 83.9 to 89.3 wt% and TiO_2 from 0.52 to 0.62 wt%), while FeO, MgO, and CaO exhibit larger variations (FeO from 1.06 to 3.78 wt%, MgO from 0.61 to 2.51 wt%, and CaO from 0.03 to 0.23 wt%).

To study the variations of major elements within individual samples and between differently colored layers, electron microprobe analyses were performed for each glass chip. The results for EPMA (average of several data points), DCP, and INAA are in good agreement with each other. Because of the good agreement of the average microprobe analyses with the other techniques, only the results of the bulk sample analyses are given in Table 1 (except for SiO_2). For a more detailed study of the chemical variations between individual layers, a microprobe profile of 0.6 mm length was measured on one Darwin glass section (distance between individual points varied between 20 and 250 μm). The variations of SiO_2, K_2O, and MgO in the profile are shown in Fig. 3a,b. The high-silica regions most probably represent lechatelierites, corresponding to depletions of all other major elements.

Correlation plots of Al_2O_3 vs. FeO or K_2O vs. FeO (not given here) show two different sections in the resulting curve: a positively correlated part with a steep slope grading into a negatively correlated curve with a less pronounced slope. The SiO_2 vs. Al_2O_3 and MgO vs. FeO plots exhibit only one trend: a negative correlation for the first pair and a positive one for the latter. A three-dimensional plot of SiO_2 vs. FeO and MgO is given in Fig. 4 and demonstrates clearly these two components. This observation can be explained by mixing at least two components with dissimilar elemental abundance ratios, with one component being rich in K_2O and SiO_2. The mixing of the silica-rich component (quartzite?) may not have been complete, leading to lechatelierite inclusions.

The color differences between individual layers are probably due to changes in the oxidation state of Fe, because no major variation of the Fe content was found in the microprobe profiles. The Fe content of the frothy white parts does not differ significantly from the colored parts of the glass. Thus the optical appearance might be due to a variable number of vesicles and differences in the oxidation state of Fe.

The trace element abundances (Table 1) also show large variations between individual samples. The concentrations of Li, Be, Sc, Rb, REE, Ta, and Th vary by a factor of less than 2; Be, Zr, Cd, Ba, Hf, and U show variations by a factor of about 2; and Cr, Mn, Co, Ni, Cu, Zn, and Sb vary by factors that are larger than 2. Some of these differences have been observed before by TAYLOR and SOLOMON (1964) and led them to divide the Darwin glasses into two distinct groups: one with average element abundances and another enriched in Cr, Ni, and Co.

4.2. *Group classification*

Since we have a more extensive data set available, we tried to verify and expand the group classification of TAYLOR and SOLOMON (1964). Statistical methods were used for discriminating the data (cluster and discriminant analysis with the elements Fe, Mg, Na, Ti, Cr, Co, and Ni, following techniques described by HOWARTH and SINDING-LARSEN, 1983, and MILLIGAN, 1980). The results of these calculations show that our glass samples form three distinct chemical groups. The groups are termed A, B, and C and can be characterized as follows: group A (56% of our samples) as average Darwin

Table 1. Major and trace element composition and group classificaton (A, B, C) of 18 Darwin glass samples

	DG01 C	DG02 B	DG03 A	DG04 A	DG05 A	DG06 A	DG07 B	DG08 A	DG09 A	DG10 C	DG11 A	DG12 A	DG13 C	DG14 B	DG15 B	DG16 C	DG17 A	DG18 A	AVG
SiO_2%	84.7	85.1	87.8	87.8	87.0	86.6	84.1	87.1	89.3	84.8	85.8	86.3	86.9	84.0	84.5	84.7	87.5	86.1	86.1
Al_2O_3%	7.66	8.04	6.75	7.19	6.77	6.59	7.63	7.44	7.00	7.50	6.83	7.21	7.25	8.20	8.47	5.79	7.17	6.90	7.25
FeO*%	2.62	3.49	1.08	1.16	2.14	2.11	3.37	1.79	1.06	2.70	2.44	2.67	1.89	3.33	3.78	2.37	1.66	2.25	2.51
MgO%	1.13	0.61	0.67	0.67	0.70	0.68	0.66	0.66	0.82	0.78	0.66	0.63	1.46	0.63	0.73	2.51	0.75	0.61	0.85
CaO%	0.06	0.03	0.04	0.11	0.04	0.03	0.03	0.09	0.11	0.06	0.08	0.07	0.18	0.07	0.12	0.23	0.10	0.08	0.09
Na	361	217	211	215	232	220	217	231	248	283	227	243	462	230	277	708	299	245	285
K_2O%	2.16	2.93	1.76	2.42	1.80	2.12	2.36	2.40	1.99	2.76	1.69	2.02	1.66	2.08	1.73	1.51	1.62	1.70	2.04
TiO_2%	0.61	0.59	0.55	0.58	0.53	0.53	0.58	0.55	0.58	0.56	0.52	0.55	0.53	0.59	0.62	0.52	0.54	0.54	0.56
Li	32	38	20	27	24	25	27	20	22	25	18	22	20	14	29	15	26	27	24
Be	0.2	0.2	0.2	0.5	0.2	0.2	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.3	0.3	0.2	0.3	0.2	0.2
Sc	7.6	8.1	6.9	7.3	7.4	7.2	8.2	7.2	6.3	6.9	6.9	6.5	6.0	7.4	7.8	6.6	7.2	6.9	7.1
Cr	103	51	54	522	56	55	52	48	65	95	50	60	151	99	83	324	77	68	86
Mn	201	54	53	32	38	42	50	43	48	202	51	46	98	251	282	207	54	54	99
Co	22.0	4.6	5.5	4.9	6.7	6.3	5.2	6.1	7.9	16.8	5.7	4.6	18.2	14.6	16.0	39.0	6.4	4.9	11.0
Ni	207	30	52	59	67	70	51	84	112	147	55	82	315	68	80	536	74	62	120
Cu	6	6	3	4	6	3	9	12	10	19	8	9	10	12	9	10	9	9	9
Zn	11	19	7	19	10	7	12	15	19	13	10	12	12	20	19	10	9	12	13
Ga	-	-	-	-	-	-	-	-	-	-	-	5	4	9	6	5	3	4	5
As	·	·	·	·	·	·	0.4	·	·	0.7	0.4	·	·	·	·	·	·	·	0.5
Rb	106	117	86	86	84	86	102	80	77	100	72	102	98	122	137	71	93	94	95
Zr	461	410	547	460	403	470	363	454	360	410	412	278	478	295	293	254	281	518	397
Cd	0.6	0.1	0.03	0.2	0.1	0.1	0.1	0.2	0.2	0.1	0.1	0.2	0.1	0.1	0.1	0.2	0.1	0.1	0.2
Sb	-	-	-	-	-	-	-	-	-	-	-	0.3	0.2	0.6	0.5	0.5	0.3	0.3	0.4
Cs	3.9	4.5	3.0	3.0	3.2	2.9	3.9	2.9	2.5	3.2	2.9	3.6	3.2	4.3	4.3	2.5	3.2	3.3	3.3
Ba	258	346	182	306	288	218	253	265	240	232	266	349	302	436	450	257	257	339	291
La	43.6	43.6	46.5	42.4	44.8	43.2	43.9	42.4	37.3	35.0	38.0	36.2	35.5	37.5	42.3	35.2	40.7	37.0	40.3
Ce	96.0	96.6	97.8	97.8	96.8	94.0	93.7	93.6	87.8	81.8	85.3	70.0	74.0	72.3	86.6	80.6	90.2	83.1	87.7
Nd	40	36	38	42	37	34	38	34	32	35	36	33	30	33	33	29	34	31	34.7
Sm	8.3	7.9	8.9	9.0	8.1	7.6	7.5	7.4	7.5	6.7	7.2	8.0	6.6	7.5	8.1	7.0	8.5	7.4	7.7
Eu	1.4	1.4	1.5	1.4	1.3	1.3	1.4	1.3	1.3	1.2	1.3	1.2	0.9	1.3	1.4	1.2	1.3	1.3	1.3
Tb	1.4	1.4	1.4	1.4	1.4	1.4	1.3	1.3	1.2	1.2	1.3	1.1	1.0	1.3	1.4	1.2	1.3	1.3	1.3
Dy	7.9	7.5	7.8	7.7	6.8	7.8	6.7	6.6	7.1	6.5	6.5	6.8	5.5	6.5	8.1	6.0	7.7	8.1	7.1
Yb	4.2	4.5	4.0	4.0	4.0	4.0	4.4	3.8	3.2	3.5	4.0	3.6	2.7	3.4	3.9	3.0	3.5	4.0	3.8
Lu	0.7	0.8	0.7	0.7	0.7	0.7	0.7	0.6	0.6	0.5	0.5	0.6	0.5	0.7	0.8	0.6	0.6	0.7	0.6
Hf	15	13	20	20	16	16	14	15	15	13	14	12	13	11	13	15	14	13	15
Ta	1.4	1.6	1.5	1.6	1.3	1.3	1.2	1.3	1.1	1.2	1.2	1.1	1.1	1.4	1.7	1.1	1.4	1.3	1.3
Th	18	18	19	19	18	17	17	17	15	15	16	14	12	15	16	16	16	15	16.3
U	3.3	3.2	2.1	5.4	2.6	3.0	2.8	2.8	1.9	2.0	2.3	2.6	1.5	3.7	3.4	3.1	2.4	2.6	2.8
(La_N/Yb_N)	7.8	7.2	8.6	7.8	8.2	8.3	7.4	8.3	7.9	6.8	6.5	6.7	8.7	7.2	6.9	7.8	7.4	6.0	7.5

Data are in ppm except where indicated in wt% *: Total Fe expressed as FeO, -: not determined, ·: data is below detection limit

glass or L(low)Fe,Al group; group B (22%) as H(high)Fe,Al group; and group C (22%) as HMg,Na group. The group classification of our samples is given in Table 1. We would like to emphasize that the present distinction between groups A and B is based on statistical analyses of our 18 samples, but more analyses may alter this classification. The three-dimensional variation plot between FeO, MgO, and Ni given in Fig. 5a shows that the glasses of groups A and B are related, with rather constant abundances of Ni and MgO. The glasses of group C are enriched in Mg, Na, Cr, Co, and Ni and are in no obvious mixing relation with the other groups. Therefore, these elements must have been supplied by a different source.

4.3. *Target rocks*

The major and trace element data listed in Table 2 demonstrate that the shales (B1 to B3-DG) and sandstones (C1-DG to C3-DG) show little chemical variation within their groups, and that element abundances are typical of upper crustal sediments. The low CaO and Na concentrations, which characterize the glasses, were also found in the target rock samples analyzed in the course of this work. CaO varies between 0.03 and 0.23 wt%, and Na between 211 and 708 ppm. The relation of target rocks to Darwin glass is discussed in more detail in Section 5.

4.4. *Anomalous element enrichments in Darwin glass*

The concentrations of Ni, Co, and Cr in group C exceed the highest abundances found in the target rocks available to us. TAYLOR and SOLOMON (1964) already noticed that these elements show abundances which are anomalous for terrestrial sediments. Subtracting an indigenous Ni contribution of about 25 ppm Ni (calculated from target rock mixing models) leaves an average excess of 48 ppm for group A and 22 ppm for group B. For samples in group C (e.g., DG8716 with an excess of 512 ppm) the enrichment can only be explained by a source different from sedimentary rocks. For Co an indigenous abundance of about 2 ppm is assumed, leading to an excess of 37 ppm Co in DG8716. Mixing of target rocks can explain the Cr concentrations in group A and B, but the Cr content of group C samples (e.g., DG8716 with 324 ppm) is also anomalously high. Estimating an indigenous contribution of 80 ppm Cr, DG 8716 contains an excess of 244 ppm Cr.

The following ratios (corrected for indigenous contributions) for DG8716 can be calculated: Ni/Co 19, Cr/Ni 0.5, Cr/Co 12. According to TUREKIAN and WEDEPOHL (1961), the ratios for shales and for sandstones, respectively, are as follows: Ni/Co 3.6; 6.7, Cr/Ni 1.3; 18, Cr/Co 4.7; 117. The Ni/Co ratio in Darwin glass is dissimilar to any terrestrial value, and therefore an extraterrestrial origin of this contam-

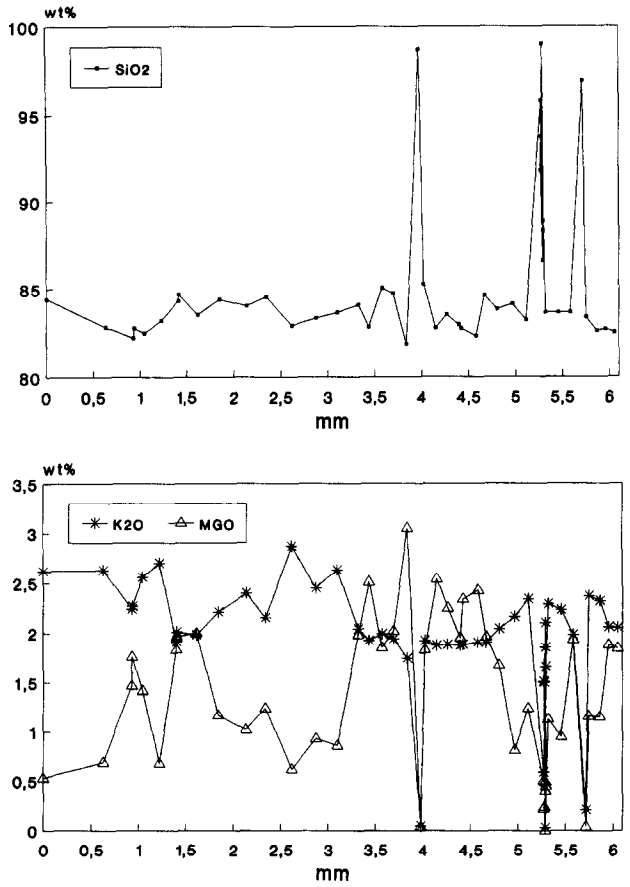


FIG. 3. Electron microprobe profile of a Darwin glass section. (a) High-silica zones are lechatelierite inclusions which correlate with the minima of MgO and K_2O contents in (b). The chemical heterogeneity of the glass in microscopical dimensions is clearly evident from these plots.

ination might be considered. MgO, Cr, Co, and Ni are positively correlated with each other (see Fig. 5b); thus, a single component may have been the source for these anomalies.

5. COMPARISON OF DARWIN GLASS WITH TARGET ROCKS

To establish the relation between country rocks from the Darwin crater and Darwin glass, we performed mixing calculations for various amounts and combinations of target rocks. The models are useful to determine the closest possible fit for the Darwin glass parent material. The results of three different mixing models are presented in Table 3. The mixing models were obtained by simply mixing different mass percentages of the target rocks; thermodynamic calculations (e.g., taking the equations of state for the different target materials into account) would complicate the models to a great extent. As demonstrated by the calculations, no perfect fit could be obtained. It can be inferred, however, that a mixture of quartzite, shales, and possibly sandstones has been the parent material of Darwin glass. The good agreement between the models and the glass chemistry is shown in Fig. 6a–d. Similar mixing models have been postulated for Australasian tektites (e.g., TAYLOR, 1962a,b; TAYLOR and KOLBE, 1964).

The following observations can be made regarding individual elements:

Major elements. The most unusual feature of Darwin glass and the target rocks is the low abundance of Ca and Na. A shale (B1-DG) has the highest CaO content of all target rocks with 0.71 wt%. The quartzite A-DG contains the least amount with 0.03 wt% CaO. A sandstone (C2-DG) has the lowest Na content (236 ppm) and another sandstone (C1-DG) the highest (1020 ppm). The glasses have CaO abundances between 0.03 and 0.23 wt% and Na contents between 211 and 708 ppm. The mixing calculations can reproduce the high concentrations of Na in glasses of group C, but give almost twice as much sodium than the average Darwin glasses of groups A and B.

TAYLOR and SOLOMON (1964) also found very low CaO and Na concentrations in Darwin glass and therefore suggested a parent material lacking plagioclase feldspar. To explain these observations we analyzed the mineral content of two samples, B2-DG and C1-DG, by X-ray diffraction. Quartz, mica, and microcline are the major components of the sediments, and no plagioclase feldspar was identified. Thus, plagioclase as a Na- and Ca-bearing component is lacking in the sediments (which may be due to weathering effects), explaining the low CaO and Na_2O abundances in the impact glass.

The K/Na ratio for Darwin glass is enhanced compared to the target rock ratios, which is due to lower Na concentrations in the glass. It has been noticed in former studies that some impact melts at other craters have higher K_2O/

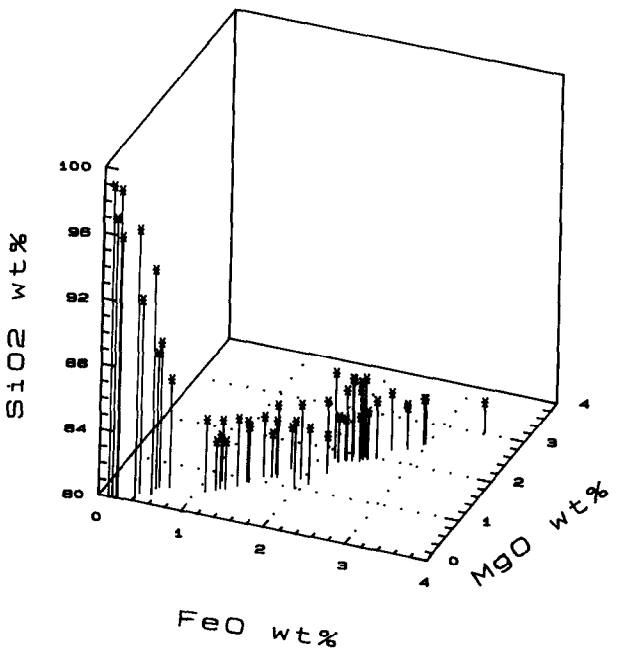


FIG. 4. This three-dimensional plot of SiO_2 vs. FeO and MgO (microprobe data of multiple points in one specimen) shows two distinct components of the glass. The points in the lower left corner of the diagram (high silica) represent the lechatelierite inclusions.

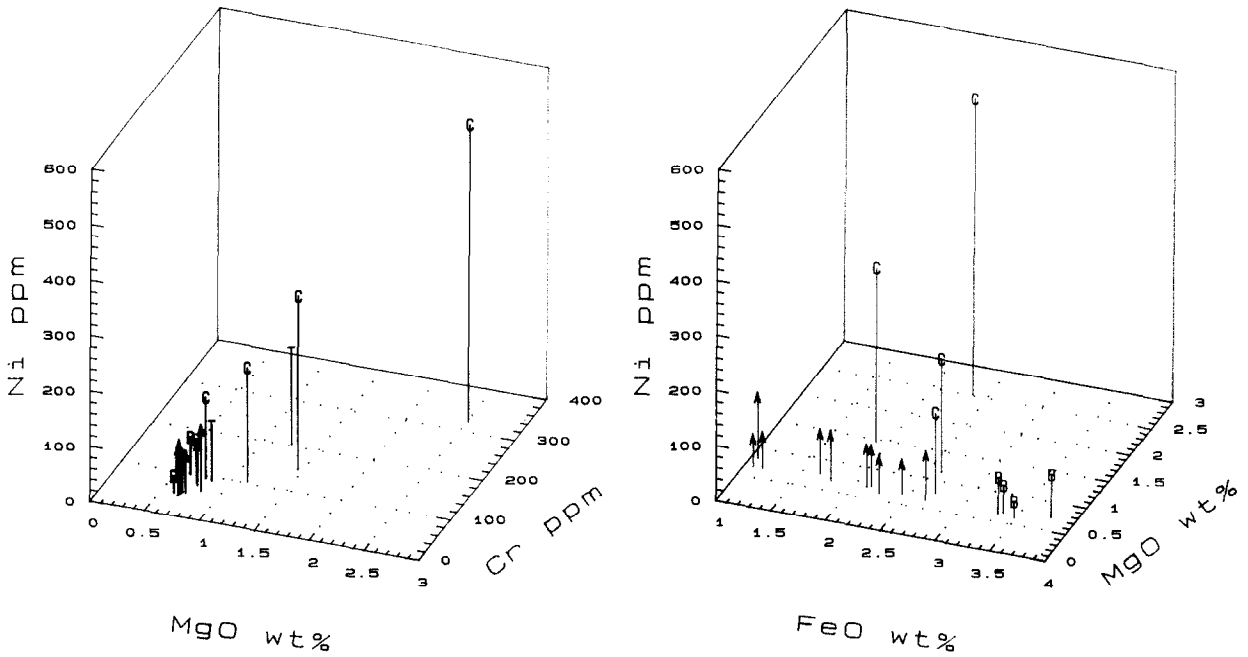


FIG. 5. (a) Three-dimensional plot of Ni vs. FeO and MgO. This is one of the plots that can be used to distinguish the three glass groups (A, B, C). Note the large variation of Ni and MgO in group C glasses. (b) This three-dimensional plot of MgO vs. Ni and Cr shows the strong positive correlation of these three elements in glasses of group C; T = glasses from TAYLOR and SOLOMON (1964).

Na_2O ratios compared to the target rocks (GRIEVE, 1987). BASILEVSKY et al. (1982) suggested that this is caused by selective elemental loss and condensation, while DENCE (1971) and GRIEVE (1978) assumed that hydrothermal alteration can explain the different elemental ratios.

Boron and fluorine. Although these two elements are not included with our analyses, they are worth discussing because some new data are available. MATTHIES and KOEBERL (1990) report an average of 11 ppm B for four samples, which is lower than the 30 ppm reported by TAYLOR and KAYE (1969), while the concentrations in the target rocks range from 19 to 64 ppm. For F, MATTHIES and KOEBERL (1990) report an average of 30 ppm F for four samples and give a F/B ratio of 2.7 (incorporating data from KOEBERL et al., 1984b). For the target rocks, an average ratio of 14.7 (four samples) was found, which is significantly higher than the ratio for Darwin glass. This is explained by MATTHIES and KOEBERL (1990) as being due to selective volatilization.

Scandium, copper, zirconium, barium, tantalum. No major discrepancies are present between the ranges of these elements in the target material and the glass; the mixing models provide a good match.

Chromium. The mixing models give slightly higher Cr abundances compared to the averages for groups A and B (62 ± 15 ppm), but the average of group C (168 ± 107 ppm) shows higher abundances, with 324 ppm Cr (DG 8716) as maximum.

Manganese. The average Mn content of all glass groups is 99 ppm, and even the average of group A (44 ppm) is slightly higher than the highest content observed in target rocks (36 ppm in B1-DG). Thus, Mn may have been introduced from a source that is different from the target rocks available for this study.

Cobalt. The Co contents of all glasses are considerably higher than concentrations in target rocks. DG8716 has the highest Co content (39 ppm) of our samples, but CHAPMAN and KEIL (1967) report values as high as 43 ppm. The averages for groups A and B are 6 and 10 ppm, respectively. Group C glasses are enriched in Co (similar to Ni and Cr) with an average of 24 ppm. The target rocks contain 2.8 ± 2.9 ppm, with a range from 0.42 to 8.3 ppm. Thus the normal mixing models are unable to explain the high Co contents in group C glasses.

Zinc. The average Zn content of the glasses is 13 ± 4 ppm (7–20 ppm) and that of the sediments is 43 ± 43 ppm (9–123 ppm). The mixing models predict higher Zn abundances, which can be explained by selective volatilization. This is similar to observations made for tektites, which have lower Zn contents (e.g., 2 ppm for australites; KOEBERL, 1986). Darwin glass and other impact glasses show smaller depletions of Zn compared to sediments than tektites, indicating a lower formation temperature.

Gallium. Gallium was analyzed in only seven glass samples. The contents range from 3 to 9 ppm, in accordance with the range of 5.6 to 10 ppm reported by TAYLOR and SOLOMON (1964). The target rocks contain more Ga than the glass, varying from 7 to 23 ppm; thus, it is reasonable to assume that a selective loss of Ga occurred during glass formation.

Rubidium. The mixing models give higher Rb contents (141–171 ppm) than the highest measured abundance in Darwin glass (137 ppm). However, the average K/Rb ratio in glass is 222, which is identical to the ratio in target rocks

Table 2. Major and trace element compositon of target rocks

	A-DG Quartzite	B1-DG	B2-DG Shales	B3-DG	C1-DG	C2-DG Sandstones	C3-DG
SiO_2 %	92.7	68.6	70.4	77.3	87.4	89.4	87.4
Al_2O_3 %	4.18	14.3	14.7	11.4	7.10	5.78	6.87
FeO* %	0.11	4.14	4.91	1.44	0.36	0.40	0.57
MgO %	0.42	3.42	3.13	1.85	0.58	0.58	0.90
CaO %	0.06	0.71	0.05	0.06	0.04	0.05	0.06
Na	458	847	398	352	1020	239	245
K_2O %	1.15	3.84	3.65	3.96	2.30	2.14	1.91
TiO_2 %	0.44	0.80	0.79	0.71	0.45	0.49	0.41
H_2O- %	0.08	0.15	0.01	0.16	0.04	0.01	0.01
L.O.I. %	0.67	4.07	2.31	3.17	1.55	1.08	1.77
Sc	3.1	11.6	11.8	9.9	7.1	5.2	6.4
Cr	86	100	94	65	41	81	72
Mn	9	36	11	21	6	7	13
Co	1.1	2.8	1.2	5.0	8.3	0.4	0.5
Ni	5	61	42	15	23	11	11
Cu	2	33	17	5	2	<2	<2
Zn	30	123	82	24	18	9	15
Ga	7	15	18	15	23	11	11
Rb	67	323	326	204	130	103	118
Zr	634	116	104	195	98	272	256
Sb	<1	1	1	2	<1	<1	1
Cs	1.4	9.1	7.2	7.1	4.0	3.3	3.5
Ba	133	195	587	461	353	279	315
La	17.0	34.4	41.8	38.8	29.4	21.3	36.1
Ce	25.7	72.8	90.4	84.5	63.1	46.4	59.4
Nd	9.8	29	34	28	27	19	21
Sm	2.0	5.5	8.2	6.4	4.6	2.9	4.7
Eu	0.4	1.0	1.4	1.1	1.0	0.5	0.9
Tb	0.5	0.7	1.5	1.0	0.7	0.6	0.8
Dy	2.7	4.6	7.0	5.4	4.2	3.7	3.9
Yb	2.2	2.5	3.4	2.7	2.8	2.3	2.7
Lu	0.4	0.5	0.6	0.5	0.5	0.4	0.4
Hf	18	9	8	11	9	9	7
Ta	1.0	1.2	1.3	1.5	1.0	1.0	1.1
Th	6.2	17.3	16.4	16.4	13.1	4.7	7.7
U	2.5	3.2	4.9	3.8	3.1	2.2	1.4

All concentrations in ppm except were marked in wt%
*: Total Fe expressed as FeO

(227). A selective loss of K and Rb (and Na) has to be considered.

Antimony. Like other volatile elements (Fig. 6), Sb is depleted in the Darwin glasses (without differences between groups) relative to the target rocks and the mixing models. The depletion is similar to that observed in tektites (KOEBERL, 1986).

Cesium. The average Cs concentration in the glasses is 3.3 ppm, which is virtually identical to the 3.6 ppm of TAYLOR and SOLOMON (1964). For the target rocks, an average of 5.1 ppm was calculated. The mixing models give Cs abundances which are slightly, but not significantly, higher than for the glass.

Thorium and uranium. The average Th and U contents of all glass groups are 16.3 and 2.8 ppm, respectively, and do not show much variation between groups. The mixing models are in good agreement with the Th contents of the glass, but give slightly higher results for U.

REEs. The chondrite normalized REE abundances for all Darwin glass samples plot within a narrow range and do not vary significantly. Figure 7a gives the range of REE abundances by plotting the samples with the highest and lowest REE contents in comparison with PAAS (post-Archean Australian sediment). All patterns have pronounced negative Eu-anomalies and are typical of post-Archean sedimentary rocks. The samples of group C show the lowest REE concentrations. Figure 7b and c gives the REE patterns of the target rocks, again in comparison with PAAS. The quartzite (A-DG) has the lowest abundances, while one of the shales (B2-DG) has higher REE contents than the glasses. Mixing calculations reproduce the shape of the patterns very well. The REE pattern of PAAS is similar to Darwin glasses regarding the absolute abundances of LREEs, but Darwin glass shows slightly higher HREE contents, which can be explained by a higher zircon contribution from the target rocks.

6. MIXING CALCULATIONS FOR ANOMALOUS ELEMENT ENRICHMENTS

None of the country rocks analyzed by TAYLOR and SOLOMON (1964) or the target rocks analyzed in the course of this work can be considered as source of enrichments of Cr, Ni, and Co that are found in Darwin glass. This implies another, non-sedimentary source, for which two possibilities can be proposed: 1) a terrestrial ultrabasic source rock; or 2) contribution from an extraterrestrial body. Ultrabasic source rocks as cause of the enrichments have been suggested by MEISEL and KOEBERL (1988). However, such rocks are not known from the crater area, and are rather unlikely to occur in the Silurian and Devonian sedimentary series of the Darwin crater, but their existence cannot be ruled out completely.

If the enrichment is of extraterrestrial origin (from the impactor), platinum group element abundances are important to identify the composition of the impacting body (e.g., MORGAN et al., 1975; PALME et al., 1981). For Darwin glass, Pt-group data are very sparse. BAEDECKER and EHMANN (1965) reported 0.2 ppb Ir in one sample, and KOEBERL (unpubl. data) found 0.02 ppb in another sample. Our INAA analyses did not show any evidence of an Ir enrichment and only an upper limit of detection of 0.2–0.6 ppb was calculated.

We performed mixing calculations aimed at reproducing the anomalous enrichments of Cr, Ni, Co, and other elements in group C glasses. A variable amount (i.e., a few percent) of terrestrial ultrabasic or meteoritic material was added to a normal glass (DG 8703) from group A (representing a homogeneous mixture of the target rocks present at the impact site, including loss of volatiles and other physico-chemical impact processes). The materials used for these calculations were a dunite, IIB and IIIB iron meteorites, an achondrite (ureilite), and a C1 chondrite. The results are given in Table 4.

By comparison with data from a typical group C glass, DG 8716, it is evident that the mixing calculations do not provide a straightforward explanation for the origin of the enrichments. Addition of ultrabasic material yields excess MgO (and FeO), but otherwise gives a good agreement for most of the minor elements. Only iron meteorites with low Ir abundances have been chosen for the mixing calculations, since there is a clear Ir deficiency in Darwin glass. The agree-

Table 3. Average data for the Darwin glass groups and comparison with the mixing models

	A	B	C	AVG	M1	M2	M3
SiO_2 %	87.1 (1.0)	84.4 (0.5)	85.3 (1.0)	86.1 (1.5)	78.6	82.8	81.1
Al_2O_3 %	7.0 (0.3)	8.1 (0.4)	7.1 (0.9)	7.3 (0.6)	10.6	8.6	9.5
FeO* %	2.0 (0.7)	3.7 (0.3)	2.5 (0.65)	2.5 (0.87)	2.9	1.9	1.3
MgO %	0.69 (0.06)	0.66 (0.05)	1.47 (0.75)	0.85 (0.46)	2.1	1.6	1.6
CaO %	0.08 (0.03)	0.06 (0.04)	0.13 (0.08)	0.09 (0.05)	0.19	0.19	0.13
Na	237 (25)	235 (28)	453 (185)	285 (123)	490	480	433
K_2O %	1.95 (0.29)	2.28 (0.51)	2.02 (0.56)	2.04 (0.40)	2.79	2.39	3.11
TiO_2 %	0.55 (0.02)	0.60 (0.01)	0.55 (0.04)	0.56 (0.03)	0.66	0.59	0.64
Sc	6.9 (0.3)	7.9 (0.4)	6.8 (0.7)	7.1 (0.6)	8.5	7.0	8.0
Cr	58 (9)	71 (24)	168 (106)	86 (65)	92	89	75
Mn	44 (8)	159 (124)	177 (53)	99 (85)	15	14	19
Co	6 (1)	10 (6)	24 (10)	11 (9)	1.4	1	4
Ni	72 (18)	57 (21)	301 (171)	120 (124)	31	24	17
Cu	7 (3)	9 (3)	11 (5)	9 (4)	14	11	7
Zn	12 (4)	18 (4)	12 (1)	13 (4)	67	55	36
Ga	4 (1)	8 (3)	5 (1)	5 (2)	13	12	13
As	0.4	0.4	0.7	0.5 (0.2)			
Rb	86 (8)	119 (14)	94 (16)	95 (18)	171	141	166
Zr	418 (90)	340 (57)	401 (102)	397 (88)	283	352	319
Cd	0.1 (0.05)	0.1 (0.02)	0.2 (0.2)	0.2 (0.1)			
Sb	0.3	0.6	0.4	0.4 (0.2)	1	1	1.6
Cs	3.0 (0.3)	4.2 (0.2)	3.2 (0.6)	3.3 (0.6)	5.5	4	5.6
Ba	271 (51)	371 (91)	262 (29)	291 (71)	342	265	336
La	40.8 (3.6)	41.8 (3.0)	37.3 (4.2)	40.3 (3.8)	35	30	37
Ce	89.6 (8.6)	87.3 (10.9)	83.1 (9.3)	87.7 (9.1)	77	64	80
Nd	35.2 (3.3)	34.8 (2.7)	33.5 (4.9)	34.7 (3.4)	28	23	26
Sm	8.0 (0.7)	7.7 (0.3)	7.2 (0.8)	7.7 (0.7)	5.2	4.1	4.9
Eu	1.3 (0.1)	1.4 (0.1)	1.2 (0.2)	1.3 (0.1)	0.9	0.7	0.9
Tb	1.3 (0.1)	1.3 (0.1)	1.2 (0.2)	1.3 (0.1)	1.0	0.8	0.9
Dy	7.3 (0.6)	7.2 (0.7)	6.5 (1.0)	7.1 (0.8)	6.4	5	5.8
Yb	3.8 (0.3)	4.1 (0.5)	3.3 (0.7)	3.8 (0.5)	2.9	3	2.7
Lu	0.6 (0.1)	0.7 (0.1)	0.6 (0.1)	0.6 (0.1)	0.5	0.5	0.5
Hf	15 (3)	13 (1)	14 (1)	15 (2)	11.3	12	13
Ta	1.3 (0.2)	1.5 (0.2)	1.2 (0.2)	1.3 (0.2)	1.2	1.1	1.5
Th	16.6 (1.6)	16.7 (1.5)	15.4 (2.6)	16.3 (1.8)	12.4	10	19
U	2.8 (1.0)	3.3 (0.4)	2.5 (0.9)	2.8 (0.9)	3.6	3.1	3.4

A: Group A (average, LFe,Al) average of 10 samples *All data in ppm, except where marked in wt%.*
B: Group B (HAl,Fe) average of 4 samples
C: Group C (HMg,Na) average of 4 samples
AVG: Average of all samples
standard deviation in ( )

M1: Mixing model 30% A-DG + 20% B1-DG + 40% B2-DG + 10% C2-DG
M2: Mixing model 40% A-DG + 20% B1-DG + 20% B2-DG + 20% C2-DG
M3: Mixing model 30% A-DG + 10% B1-DG + 60% B3-DG

ment between Darwin glass and iron meteorite mixtures is not very good. The Cr and Mn contents are too low, and the Fe, Co, and Ni contents are much too high compared with Darwin glass.

An achondritic projectile (e.g., the ureilite used in the mixing calculations) provides a better fit than iron meteorites. In this particular case, the Co and Ni abundances of the mixture are lower, but still at the same magnitude. The model produces Ir abundances of about 30 ppb, which is at least two orders of magnitude above the observed abundances. The same problem is even more evident for contamination by a chondritic bolide. Mixing with 5% C1 material gives a good agreement for Cr, Mn, and Co, too high Fe and Ni, but again, Ir abundances that are in excess by more than two orders of magnitude.

Thus, as far as the question of the composition of the projectile or the exact origin of the element enrichments is concerned, we are left with a dilemma. Stony meteorites (chondrites and achondrites) provide too much Ir, although some other elements can be fitted very well. Furthermore, PALME et al. (1981) state that most small impact craters are produced by iron or stony-iron projectiles because of atmospheric breakup of the more fragile stony meteorites. But iron meteorite mixtures give no good agreement at all (a similar problem has been noted by O'KEEFE, 1987, for two other craters). A dunite mixture would provide a better fit, except for a slight Fe excess (which is nevertheless smaller than for all other mixtures), but ultrabasic rocks are not known from the crater area. It seems that either some ultrabasic rock or an achondrite (with slightly different absolute abundances than the ones used for our calculations) would provide a good agreement. However, a cometary impact cannot be ruled out either, as already suggested for other impacts (e.g., SCHMITT, 1989). For example, Halley dust has lower Fe and Ni abundances

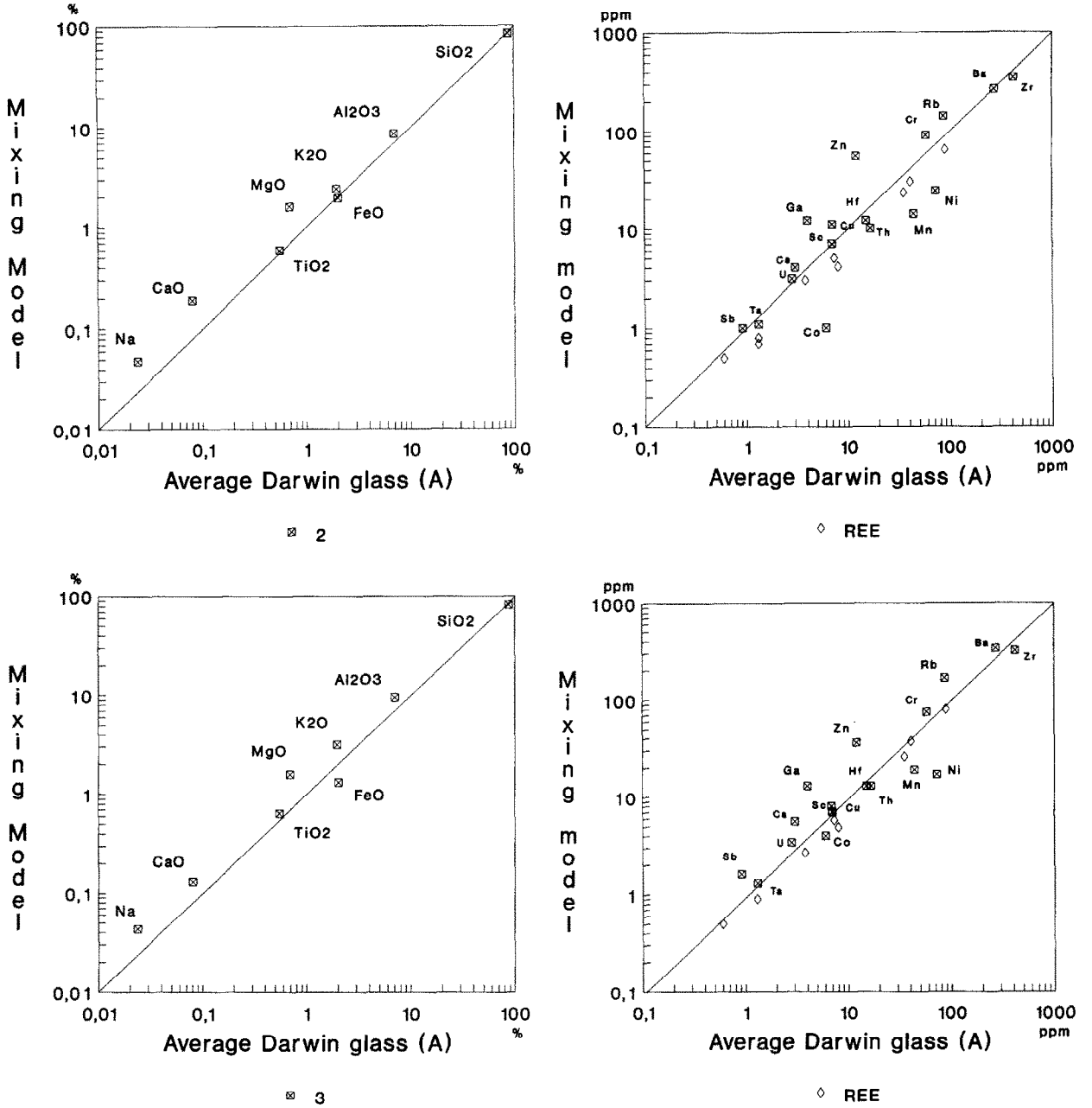


FIG. 6. Correlation plots of data from the mixing models (using target rock data, see Table 3) vs. Darwin glass: (a) major and (b) trace elements of model M2 vs. the average of glass group A (average Darwin glass group); (c) major and (d) trace elements of model M3 vs. the average of the glass group A. A very good fit is evident for most elements, with the exception of some volatile elements which are lost during the impact.

than Cl-chondrites, but no Ir data is available (JESSBERGER et al., 1988).

7. CONCLUSIONS

From the data and discussions given in this paper the following conclusions can be drawn:

1. The chemical composition of 18 Darwin glass samples has been studied for major and trace elements, and is in agreement with an origin from terrestrial sediments during an impact.
2. By statistical analysis of our chemical data, we have been able to identify two closely related groups of Darwin glass (A: average Darwin glass or LFe,Al group; B: HFe,Al group); and a third group (C: HMg,Na group), which shows enrichments of Cr, Ni, and Co.
3. Analyses of target rocks from the Darwin crater and mixing calculations show that Darwin glass can be formed from the local target rocks. A mixture of 30% quartzite, 60%

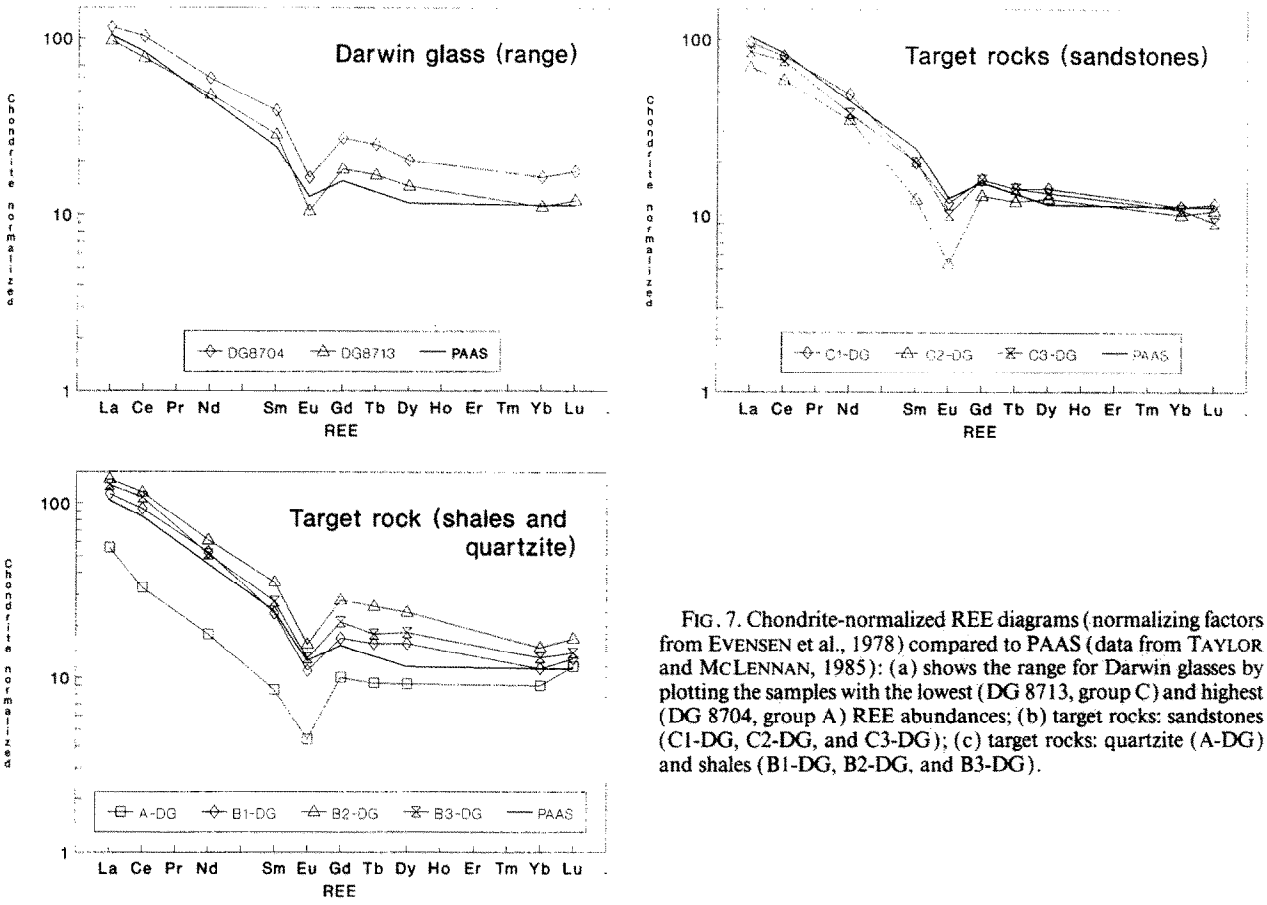


FIG. 7. Chondrite-normalized REE diagrams (normalizing factors from EVENSEN et al., 1978) compared to PAAS (data from TAYLOR and MCLENNAN, 1985): (a) shows the range for Darwin glasses by plotting the samples with the lowest (DG 8713, group C) and highest (DG 8704, group A) REE abundances; (b) target rocks: sandstones (C1-DG, C2-DG, and C3-DG); (c) target rocks: quartzite (A-DG) and shales (B1-DG, B2-DG, and B3-DG).

shale B3-DG, and 10% shale B1-DG provides the best fit for the parent material of group A and B glasses. Most elemental abundances (major and trace) are in good agreement between the mixing model and the average glass composition, but there are differences in Si, Al, and Fe. This may be due to the limited variety of target rock samples available for this study, and other (similar) rocks in the impact area may exist.

Table 4. Mixing model for anomalous element enrichments

								DG8703 +					
	DG8703	SikhAlin IIB	Acuna IIIB	AguaBl IIIB	RC027 Ureilite	CI Chondr	Dunite	IIB 2%	IIIB 5%	Ureilite 5%	CI 5%	Dunite 8%	DG8716
Al_2O_3 (%)	7.66	-	-	-	0.39	2.43	1.82	7.51	7.28	7.30	7.40	7.20	5.79
FeO (%)	2.62	124	113	117	20.8	34	12.61	5.1	8.14	3.53	4.2	3.42	2.37
MgO (%)	1.13	-	-	-	36.6	23.4	37.9	1.11	1.07	2.90	2.24	4.07	2.51
CaO (%)	0.06	-	-	-	1.2	1.94	1.01	0.06	0.06	0.12	0.15	0.14	0.23
Na (ppm)	361	-	-	-	207	7790	1485	354	343	353	732	451	708
K_2O (%)	2.16	-	-	-	0.002	0.11	-	2.12	2.05	2.05	2.06	1.99	1.51
TiO_2 (%)	0.61	-	-	-	0.2	0.11	0.09	0.60	0.58	0.59	0.59	0.57	0.54
Cr (ppm)	54	7	13	33	5280	3500	3500	53	52	315	226	330	324
Mn (ppm)	53	170	-	-	3000	2700	5650	55	51	200	186	501	207
Co (ppm)	5.5	4600	5600	5210	175	765	119	98	285	14	43	15	39
Ni (ppm)	52	5.9%	10.2%	8.44%	1830	1.51%	3000	1231	5149	141	804	288	536
Ir (ppb)	<0.2	27	24	416	580	975	1	0.74	1.4	29	49	0.1	<0.2

References: *Sikhote Alin, IIAB iron meteorite, from WEINKE et al. (1986)*
Acuna and Agua Blanca, IIIAB iron meteorites, from WASSON et al. (1989)
Ureilite RC027 from GOODRICH et al. (1987), except Mn from MASON (1979)
CI-Chondrite from TAYLOR (1982)
Dunite from SCHARBERT (1984), STUBER and GOLES (1967), and GOVINDARAJU (1984)

4. There is evidence for loss of the volatile elements Ga, Zn, F, and B during the impact event, which is expected because of the high formation temperature. This can be explained by selective volatilization of these elements from the impact melt, similar to observations made for tektites.
5. The elements Na, K, Rb, and Cs show lower abundances in the glass than in the target rocks. This is also evidence for a selective volatilization of the elements, but it is interesting to note that Cs shows the least depletion.
6. While the compositions of group A and B glasses can in general be reproduced by mixing of local country rocks, the absolute abundances of Ni and Co and also the Ni/Co, Cr/Ni ratios in glasses of group C are anomalous and cannot be explained by contributions from the normal (sedimentary) target rocks. Other sources, such as contamination by ultrabasic rocks, or from the impacting body, have to be considered.

We have performed mixing calculations by adding a few percent of ultrabasic or meteoritic material to Darwin glass of average composition to reproduce the Cr, Ni, Mn, and Co enrichments. Iron meteorites provide the least acceptable fit, while chondritic contamination would result in a much higher Ir concentration than actually observed. Better agreements are found for an ultrabasic contribution, but no such rocks are known from the crater area, or for an achondritic projectile; but here again an Ir excess is present. Further investigations are clearly necessary to obtain conclusive chemical data to identify the projectile.

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